

NIH Sample — Specific Aims & Research Strategy

Illustrative R01-style application excerpt

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Specific Aims

Hypertension is markedly more prevalent and less well controlled in historically underserved urban communities, yet most digital self-management tools are neither designed with these communities nor validated in them. **The long-term goal** of this research is to reduce disparities in blood-pressure control through community-designed digital health tools. **The overall objective** of this application is to develop and test a culturally tailored, mobile self-management intervention for adults with uncontrolled hypertension in Baltimore. **Our central hypothesis** is that a community-co-designed intervention will improve blood-pressure control and medication adherence relative to usual care. Guided by strong preliminary data from our community advisory board, we will test this hypothesis through three aims:

Aim 1. Co-design and refine the intervention with patients and community health workers. *Working hypothesis:* structured co-design yields features that map to known adherence barriers. We will use iterative design sessions and usability testing to finalize the intervention.

Aim 2. Determine efficacy on blood-pressure control in a randomized pilot trial (N = 200). *Working hypothesis:* the intervention reduces systolic blood pressure by ≥ 6 mmHg at 6 months versus usual care. Primary outcome: change in systolic BP at 6 months.

Aim 3. Identify mediators and implementation determinants using mixed methods. *Working hypothesis:* medication adherence and self-efficacy mediate the BP effect, and specific clinic-level factors predict adoption.

Expected outcomes and impact. This work will produce a validated, community-designed intervention and the implementation knowledge needed to scale it—an expected positive impact on a condition that drives substantial, preventable cardiovascular morbidity in underserved communities.

Research Strategy

Significance

Uncontrolled hypertension is a leading, modifiable contributor to cardiovascular disease, and the burden falls disproportionately on the communities this project serves. Existing digital tools show promise but are rarely co-designed with or validated in these populations, limiting both efficacy and adoption. By grounding design in the community and rigorously testing efficacy, this project addresses an important problem that, if solved, would shift how self-management tools for chronic disease are developed and evaluated.

Innovation

The project is innovative in three ways: it applies a structured community co-design method to a chronic-disease self-management tool rather than retrofitting an existing app; it pairs an efficacy trial

with an embedded implementation-science evaluation so that adoption is studied from the start; and it centers community health workers as designers and deliverers, not just recruiters.

Approach

Aim 1 (co-design). We will conduct six iterative design sessions with patients and community health workers, followed by think-aloud usability testing with 15 participants, refining the intervention to a locked design. **Aim 2 (pilot RCT).** We will randomize 200 adults with uncontrolled hypertension 1:1 to the intervention or usual care, measuring blood pressure with validated automated devices at baseline, 3, and 6 months; the primary analysis is an intention-to-treat comparison of 6-month systolic BP change. **Aim 3 (mediators and implementation).** Pre-specified mediation analyses will test adherence and self-efficacy as mediators, and qualitative interviews with clinic staff, analyzed with an implementation framework, will identify adoption determinants. **Potential pitfalls and alternatives.** If recruitment lags, we will add a second clinic site already engaged with our team; if adherence measurement by self-report proves unreliable, we will incorporate pharmacy-refill data.